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Ji Yuan Feng

EDUCATION

Stanford University, Robotics and Autonomous System Graduate Certificate **2024 - Present**

Cumulative Average: 3.6 GPA

Courses: CS 237A, CS 237B Principles of Robot Autonomy I, II, EE227 Robot Perception

University of Waterloo, Bachelor of Applied Science, With Distinction **2018 - 2023**

Mechatronics Engineering, Artificial Intelligence Option, Honors, Co-operative Program

Cumulative Average: 85.04% (3.7 GPA)

Clubs: SAE AutoDrive Challenge (Level 4 Autonomous Vehicle), Waterloo (Hyperloop)

Scholarships: President's Scholarship, Sanford Flaming Foundation Award, Term Dean's Honor List

PUBLICATIONS

J. Y. Feng, A. Kaufman, "Learn OpenUSD: Robotics Best Practices," in SIGGRAPH Labs '25, Article No.: 1, Page 1, doi: 10.1145/3721251.3736528

J. Y. Feng, R. Chadha, "Digital Twin Robotics: Immersive Software-in-the-Loop Testing with OpenUSD, Isaac Sim, and ROS," in SIGGRAPH Labs '25, Article No.: 4, Page 1, August 2025, doi: 10.1145/3721251.3734066

NVIDIA, "Isaac Lab: A GPU-Accelerated Simulation Framework for Multi-Modal Robot Learning," White Paper, NVIDIA Corporation, 2025. [Online]. Available:

https://research.nvidia.com/publication/2025-09_isaac-lab-gpu-accelerated-simulation-framework-multi-modal-robot-learning, (Contributor **J. Y. Feng**)

N. Abdulazeem, N. Sichert, **J. Y. Feng** and Y. Hu, "Quantifying Human Mental State in Interactive pHRI: Maintaining Balancing," in IEEE Robotics and Automation Letters, vol. 10, no. 3, pp. 2958-2965, March 2025, doi: 10.1109/LRA.2025.3539103.

RESEARCH

Research Assistant, Active Interactive Robotics Lab **01/2023 – 04/2023**

PI: Professor Yue Hu

- Brainstormed and assisted with conducting experiments to analyze the effect of active vs passive robot behavior on user's cognition during balancing tasks
- Developed scripts to track participant's pose and worked with the Sawyer robot using ROS
- Experimented with devices to simulate loss of balance and sensors to measure participant stress

WORK EXPERIENCES

Robotics Systems Engineer, NVIDIA Corporation **06/2023 – Present**

- Developed sim to sim transfer pipeline for evaluating reinforcement learning policies across different simulated environments in Isaac Sim, and built RL based locomotion and manipulations examples for humanoids, quadrupeds and grippers, presented at SIGGRAPH for 100+ attendees

- Unified sim ready digital twin standards by developing URDF and MJCF to USD importer for Isaac Sim, with modularized assets for data collection, simulations, and training
- Created a sim ready digital twin library of over 100 humanoids, quadrupeds, manipulators, grippers, and mobile robots for training and deployment in Isaac Sim and Isaac Lab
- Developed end to end workflows for building robotics arm digital twins and controller using inverse kinematics, presented at NVIDIA GTC and SIGGRAPH for 200+ attendees
- Mentored three interns with curated projects such as Ackermann controller, policy evaluator

Vehicle Firmware Developer Intern, Tesla Inc

05/2022 – 08/2022

- Designed “Semi Lighting Test” for controlling 19 unique light groups across 4 controllers in C for Tesla Semi truck pre-trip lighting inspection, featured at the *Tesla Semi Delivery Event*
- Set up simulations for the Tesla Semi lighting controllers and created software in a loop test using Python and drafted vehicle technical requirements on Polarion
- Brought up light fault detections and alerts for Tesla Semi tractor and trailer lights with C via CAN
- Developed self-tests for checking vehicle mirrors, latches, horns in C to speed up factory verification

Robotics Simulation Developer Intern, NVIDIA Corporation

01/2022 – 04/2022

- Developed Quadruped extension for simulating the Unitree A1 and Go1 quadrupeds in Isaac Sim
- Connected quadruped gait controllers to the model with Python and created ROS and ROS 2 stereo vision SLAM examples in Isaac Sim
- Created Isaac Sensor extension for simulating physics-based sensors, created USD schema for sensors which enables visualization and modification of sensor parameters using the GUI at runtime

Embedded Systems Developer Intern, Ecobee Inc

05/2021 – 08/2021

- Built display testing pipeline using C++, OpenCV to detect discoloration, misalignment for the Ecobee smart thermostat, capable of detecting color disparity and alignment offsets of 0.5mm

Teaching Assistant, University of Waterloo

09/2020 – 12/2020

- Hosted labs and tutorials for ECE 150 Intro to C++ course for 400+ students, designed examinations.

Innovation Intern, Khazanah America Incorporated

01/2020 – 04/2020

- Developed vision-based control method for drones using OpenCV to track and follow Aruco marker.

Software Developer Intern, Ford Motor Company

05/2019 – 08/2019

- Developed scripts for automatic scheduling and testing of vehicle chips on Jenkins and TestStand

PROJECTS

Cloud Compute Project

2024

- Integrated Ollama and OpenWebUI to host local large language models, accessible publicly
- Expanded server capabilities with cloud storage, image browser, virtual machines, and offsite backup

Multi-Robots Mapping Application (Capstone Project)

2023

PI: Professor Sanjeev Bedi

- Developed scalable SLAM and ROS navigation pipeline for multi-robots SLAM and search task using a shared server for collaborative goal assignment and global map sharing in a group of four

Local Planning Developer, SAE AutoDrive Challenge (Level 4 Autonomous Vehicle)

2021

- Integrated MPC controller with the self-driving stack using ROS and built double lane change functionalities to avoid obstacles, verified algorithm using Carla Simulator